# Harshith Kantamneni

kantamneniharshith@gmail.com | +1 (414) 916-5799 | linkedin.com/in/hk4231 | github.com/harshithkantamneni | harshithkantamneni.github.io
Milwaukee, WI

## Summary

M.S. ECE engineer across the full AI-systems stack: optimizes CUDA/Triton kernels and profiles LLM inference at the bottom; designs agent infrastructure, retrieval, and evaluation on top. Shipped kernel-fusion and GPU-profiling work for LLM workloads, an MCP memory-and-retrieval layer for coding agents, and autonomous multi-agent labs with evaluation and governance.

## Skills

- **GPU & Performance:** CUDA, Triton (kernel DSL), PyTorch, kernel fusion, mixed precision (fp16), NVIDIA Nsight Systems/Compute, GPU performance profiling, roofline/memory-bandwidth analysis, quantization, gem5, McPAT
- **AI Systems:** LLM inference profiling, FlashAttention/SDPA, KV-cache decode, multi-agent orchestration, MCP server design, Claude Code / Agent SDK, agent evaluation, governance/anti-forgery
- **Retrieval & Memory:** Hybrid search, reciprocal-rank fusion, cross-encoder reranking, sqlite-vec, SQLite FTS5, knowledge graphs
- **Languages & Infra:** Python, C++, CUDA, Docker, CI/CD, Slurm

## Education

**University of Wisconsin–Madison** — *Madison, USA*
*M.S. Electrical and Computer Engineering* — Sep 2024 – Dec 2025

- Relevant Coursework: High Performance Computing, Advanced Computer Architecture, Machine Learning, Fault-Tolerant Computing

**Vellore Institute of Technology** — *Amaravati, India*
*B.Tech in Electronics and Communication Engineering* — 2020 – 2024

## Projects

**Triton vs cuBLAS LLM Kernel Benchmarking** — Dec 2025 – Present
*Triton, CUDA, PyTorch, NVIDIA Nsight* — GitHub

- Fused linear+bias+GeLU into one **Triton** kernel for **up to 1.73**× lower p50 latency than **cuBLAS** on small-batch FFN projections, removing an intermediate **HBM** round-trip and a kernel launch.
- Benchmarked **76 LLM-shaped** GEMM kernels across Triton and cuBLAS on an **A100** (p50/p90 latency, throughput, bandwidth, tail jitter) to map where fusion wins; the tiled Triton GEMM sustains **215 TFLOP/s**, **69% of fp16 tensor-core peak**.

**Edge LLM GPU Profiling on Jetson Orin Nano** — Feb 2026
*CUDA, NVIDIA Nsight Systems, PyTorch, FlashAttention* — GitHub

- Profiled **attention** and **KV-cache decode** on a Jetson Orin Nano (edge GPU for local LLM-agent deployment), sustaining **∼15.4K tokens/sec** single-stream FlashAttention decode and **∼7.5 TFLOP/s** peak fp16 GEMM.
- Measured fused **SDPA/FlashAttention** at ∼**5.6**× over naive attention with Nsight Systems and traced a per-step decode kernel-launch-overhead pattern.

### bert: Long-Context Memory & Signed-Proof Layer (MCP) May 2026 – Present
*Python, MCP, sqlite-vec, sentence-transformers* GitHub

- Built **bert**, a local **MCP server** (JSON-RPC 2.0/stdio) that gives coding hosts like Claude Code, Cursor, and Codex per-project **long-context memory** via hybrid retrieval (dense **sqlite-vec** + **BM25** fused by **reciprocal-rank fusion**, bge cross-encoder rerank), scoring **0.684 nDCG@10** on BEIR scifact, above the published BM25 baseline.
- Root-caused and fixed a silently-broken retriever (mismatched dict keys zeroed the dense fusion signal; a 240-char truncation dropped answer spans), lifting accuracy from **0.10 to 0.85** and recall@10 from **0.13 to 0.78** on a span-validated gold set.
- Demonstrated that retrieval is the only viable path once a project exceeds the context window: on a **3M-token** corpus over a **1M** window, full-context is infeasible and truncation scores **0.00**, while bert holds **0.75** accuracy at a flat ∼**3.3K** input tokens.

### Autonomous ML Research Lab (Multi-Agent) Mar 2026 – Present
*Python, C17, Apple Metal, Claude API (Opus 4.8)* GitHub

- Design and operate a **31-agent** autonomous research lab on an 18 GB laptop (no cloud) whose agents build and test a from-scratch **C17/Metal training engine** (MoE transformer, 4-bit QAT, hand-written kernels, no ML frameworks) and run pre-registered ML research end to end.
- Hardened the lab with an integrity and governance layer: pre-registration, phase gates, and **anti-forgery** signature verification added after a governance agent forged 5 approvals, with **no recurrence over 23 sessions**.
- Engineered the lab's retrieval substrate: a **4-layer hybrid memory** over **62K chunks** (graph PageRank + sqlite-vec dense + BM25, fused via reciprocal-rank fusion and cross-encoder reranking), warm-served at ∼**32 ms**.

### HIVE: Autonomous Product Org (Rust + libp2p) Mar 2026 – Present
*Rust, libp2p, Python, Claude Code*

- Founded and operate an autonomous **45-role** multi-agent product organization (Director, Evaluator, Red Team) that has run ∼**203 cycles** building a privacy-first reasoning product; authored its governance substrate from scratch: operator-signed values, a 3-tier reversibility-based decision protocol, and JSON inter-agent handoff schemas.
- Directed agents to build a **66K-LoC Rust** system: an alpha-network **Rete** engine with justification-based truth-maintenance and MYCIN confidence scoring, a 21-module **differential-privacy** pipeline (Rényi-DP budgeting, k-anonymity, local DP), and a **libp2p** P2P layer (Kademlia DHT, Gossipsub, Dandelion++), gated by **1,300+ tests** and cargo-audit/pip-audit CVE scanning.

### ML-Guided CUDA Kernel Configuration 2025
*Python, PyTorch, CUDA, Slurm* GitHub

- Trained a PyTorch MLP runtime surrogate (12 GEMM/launch features) that predicts CUDA kernel execution time at **R2 = 0.96** (MAE 0.018 ms), replacing exhaustive autotuning with argmin-over-grid config selection.
- Validated predicted block/grid configs to **within ∼3% of the exhaustively-measured optimum** (median across workloads), with benchmarking and CUDA event-timing automated on Slurm.

### MI300X Memory-System Characterization & gem5 Modeling 2025
*HIP, gem5, McPAT, Python*

- Characterized the **AMD MI300X** memory hierarchy with HIP microbenchmarks (vL1D ∼130, L2 ∼230, HBM ∼630–760-cycle latencies; **4.6–4.8 TB/s** write bandwidth) and calibrated full-system **gem5** to **within 5% MAPE** of measured vL1D latency, down from 89%.
- Compared in-order and out-of-order CPU models in gem5 across SAXPY/DAXPY/IAXPY for CPI, latency, and power, with **McPAT** stats-to-power extraction.